Air University - Islamabad
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Lab Task 02

Full Stack Programming (B)

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Lab Tasks

1. Task 1: Dynamic Quiz:

Code:

```
<div id="quiz-container">
   <h3>Quick Quiz</h3>
   <div id="q1-div">
      <p>1. What is 5 + 5?</p>
      <input type="number" id="ans1">
   </div>
   <div id="q2-div">
      <p>2. What is the capital of France?</p>
      <input type="text" id="ans2">
   </div>
   <div id="q3-div">
      <p>3. Is JS a programming language? (yes/no)</p>
      <input type="text" id="ans3">
   </div>
   <br>
   <button onclick="checkQuiz()">Submit</button>
   <button onclick="resetQuiz()">Reset</button>
   <div id="quiz-result"></div>
</div>

<script>
   const c1 = 10;
   const c2 = "paris";
   const c3 = "yes";

   function checkQuiz() {
      let score = 0;
      let a1 = document.getElementById("ans1").value;
      let a2 = document.getElementById("ans2").value.toLowerCase();
      let a3 = document.getElementById("ans3").value.toLowerCase();

      if (a1 == c1) score++;
      if (a2 == c2) score++;
      if (a3 == c3) score++;

      let msg = "";
      if (score === 3) msg = "Excellent!";
      else if (score > 0) msg = "Good job!";
```

```
    else msg = "Try again!";

    document.getElementById("quiz-result").innerHTML = "Score: " + score + "/3 - " + msg;
  }

  function resetQuiz() {
    document.getElementById("ans1").value = "";
    document.getElementById("ans2").value = "";
    document.getElementById("ans3").value = "";
    document.getElementById("quiz-result").innerHTML = "";
  }
</script>
```

Output:

Quick Quiz

1. What is 5 + 5?

9

2. What is the capital of France?

Paris

3. Is JS a programming language? (yes/no)

yes

Submit Reset

Score: 2/3 - Good job!

2. Task 2: Interactive Calculator

Code:

```
<div id="calc-container">
  <input type="number" id="num1">
  <select id="op">
    <option value="+">+</option>
    <option value="-">-</option>
    <option value="*">*</option>
    <option value="/">/</option>
  </select>
  <input type="number" id="num2">
  <button onclick="calculate()">=</button>
  <div id="calc-res" style="padding: 10px; margin-top: 5px; width: 100px;"></div>
</div>

<script>
  function calculate() {
    let n1 = parseFloat(document.getElementById("num1").value);
    let n2 = parseFloat(document.getElementById("num2").value);
    let op = document.getElementById("op").value;
    let resBox = document.getElementById("calc-res");
    let res = 0;

    if (isNaN(n1) || isNaN(n2)) {
      resBox.innerHTML = "Enter numbers";
      return;
    }

    if (op === "+") res = n1 + n2;
    else if (op === "-") res = n1 - n2;
    else if (op === "*") res = n1 * n2;
    else if (op === "/") {
      if (n2 === 0) {
        resBox.innerHTML = "Div by 0!";
        return;
      }
      res = n1 / n2;
    }

    resBox.innerHTML = res;
    resBox.style.backgroundColor = res >= 0 ? "lightgreen" : "lightcoral";
  }
</script>
```

Output:

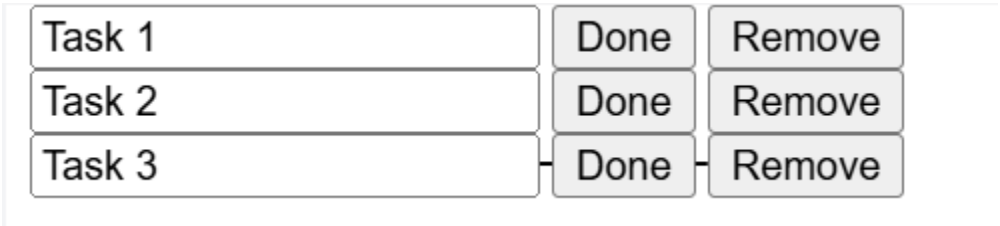
3. Task 3: Simple To-Do List

Code:

```
<div id="todo-container">
   <div class="task-item"><input type="text" value="Task 1" id="t1"> <button onclick="done(0)">Done</button>
<button onclick="rem(0)">Remove</button></div>
   <div class="task-item"><input type="text" value="Task 2" id="t2"> <button onclick="done(1)">Done</button>
<button onclick="rem(1)">Remove</button></div>
   <div class="task-item"><input type="text" value="Task 3" id="t3"> <button onclick="done(2)">Done</button>
<button onclick="rem(2)">Remove</button></div>
</div>

<script>
   function done(idx) {
      let items = document.querySelectorAll(".task-item");
      for(let i=0; i<items.length; i++) {
         if(i === idx) {
            items[i].style.textDecoration = "line-through";
         }
      }
   }
   function rem(idx) {
      let items = document.querySelectorAll(".task-item");
      items[idx].style.display = "none";
   }
</script>
```

Output:



4. Task 4: Color Box Generator

Code:

```
<div>
  <input type="text" id="c1"> <button onclick="addBox('c1')">Add</button><br>
  <input type="text" id="c2"> <button onclick="addBox('c2')">Add</button><br>
  <input type="text" id="c3"> <button onclick="addBox('c3')">Add</button><br>
  <button onclick="clearBoxes()">Clear All</button>
  <div id="box-area"></div>
  <p id="browser-info"></p>
</div>

<script>
  function addBox(id) {
    let col = document.getElementById(id).value;
    let area = document.getElementById("box-area");
    let box = document.createElement("div");
    box.style.width = "50px";
    box.style.height = "50px";
    box.style.backgroundColor = col;
    box.style.display = "inline-block";
    box.style.margin = "5px";
    area.appendChild(box);
    document.getElementById("browser-info").innerHTML = "W: " + window.innerWidth + " H: " + window.innerHeight;
  }
  function clearBoxes() {
    document.getElementById("box-area").innerHTML = "";
  }
</script>
```

Output:

W: 575 H: 451

5. Task 5: Form Validation

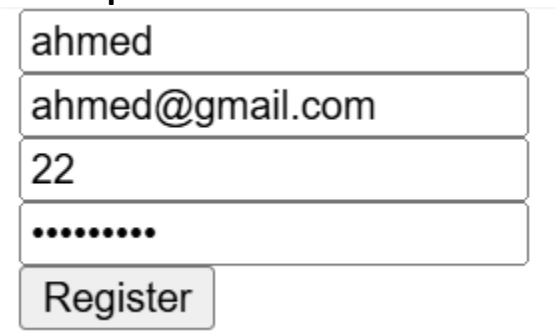
Code:

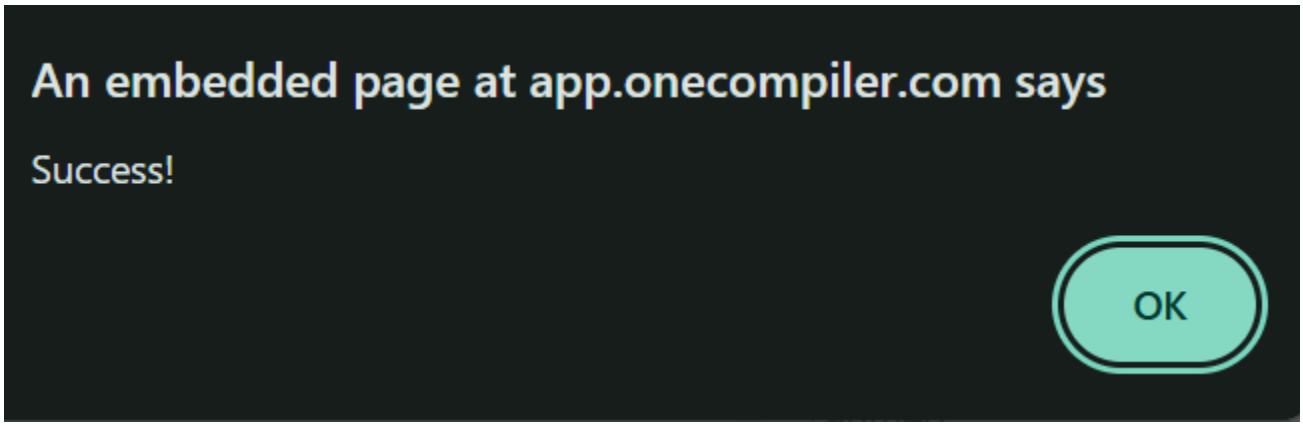
```
<form id="regForm" onsubmit="return validate(event)">
   <input type="text" id="name" placeholder="Name"><span id="e1"></span><br>
   <input type="text" id="email" placeholder="Email"><span id="e2"></span><br>
   <input type="number" id="age" placeholder="Age"><span id="e3"></span><br>
   <input type="password" id="pass" placeholder="Password"><span id="e4"></span><br>
   <button type="submit">Register</button>
</form>

<script>
   function validate(e) {
      e.preventDefault();
      let name = document.getElementById("name").value;
      let email = document.getElementById("email").value;
      let age = parseInt(document.getElementById("age").value);
      let pass = document.getElementById("pass").value;
      let valid = true;
      document.getElementById("e1").innerHTML = "";
      document.getElementById("e2").innerHTML = "";
      document.getElementById("e3").innerHTML = "";
      document.getElementById("e4").innerHTML = "";
      if (name === "") { document.getElementById("e1").innerHTML = " Name empty"; valid = false; }
      if (!email.includes("@")) { document.getElementById("e2").innerHTML = " Invalid Email"; valid = false; }
      if (age < 18 || age > 60 || isNaN(age)) { document.getElementById("e3").innerHTML = " Age 18-60"; valid = false; }
      if (pass.length < 6) { document.getElementById("e4").innerHTML = " Min 6 chars"; valid = false; }

      if (valid) {
         if (confirm("Confirm Submission?")) {
            alert("Success!");
         }
      }
      return valid;
   }
</script>
```

Output:




An embedded page at app.onecompiler.com says
Success!
OK